

Google Cloud

PRESENTS

Agentic AI Day

Build the next generation of intelligent agents

Powered by **H2S**
HACK2SKILL

Team Details

- a. **Team name: MALAAI (Machine Learning And Artificial Intelligence)**
- b. **Team leader name: Khusbu Rai**
- c. **Problem statement: Improving safety at large public events**

• Problem Identified :

1. Lack of real-time crowd monitoring is a major cause of stampedes at large gatherings
2. Delayed responses and absence of live risk insights often lead to chaos and fatalities

• Why It Matters :

3. Over 7,000 deaths globally and 1,100+ in India due to crowd-related disasters in the last 20 years

• Our Solution :

4. Detects crowd density and classifies zones (Low/Medium/High) in real time
5. Displays live heatmaps for visual risk tracking
6. Identifies missing persons via facial recognition (FaceNet + DeepFace)
7. Suggests crowd control actions using Google Gemini
8. Stores data and integrates easily with existing CCTV systems

• What Makes It Unique :

9. Goes beyond alerts — it predicts and guides real-time action
10. Combines CV, facial recognition, and AI in one dashboard
11. Scalable for use at public events, religious gatherings, and smart cities



Opportunities



• How different is it from any of the other existing ideas?

1. Traditional systems only record or detect motion
2. Our system understands crowd behavior and predicts risk in real time
3. Combines detection, risk analysis, and action in one unified dashboard



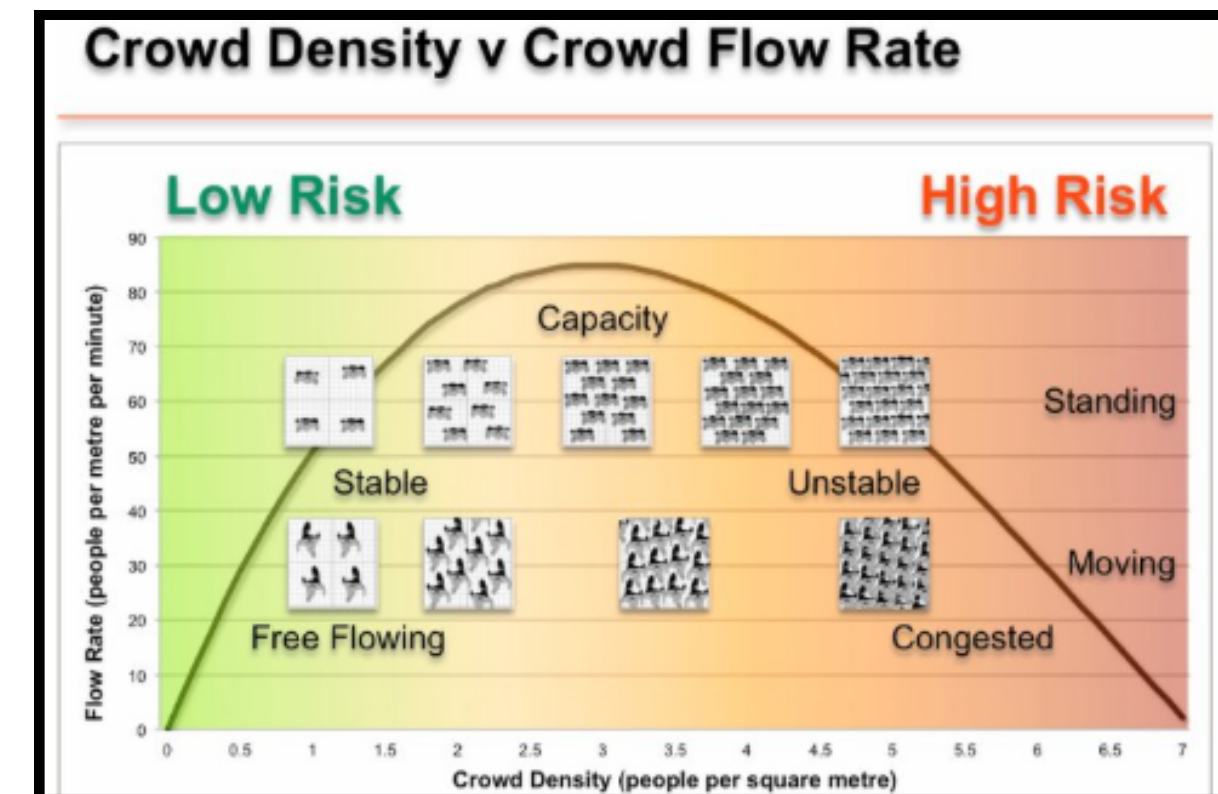
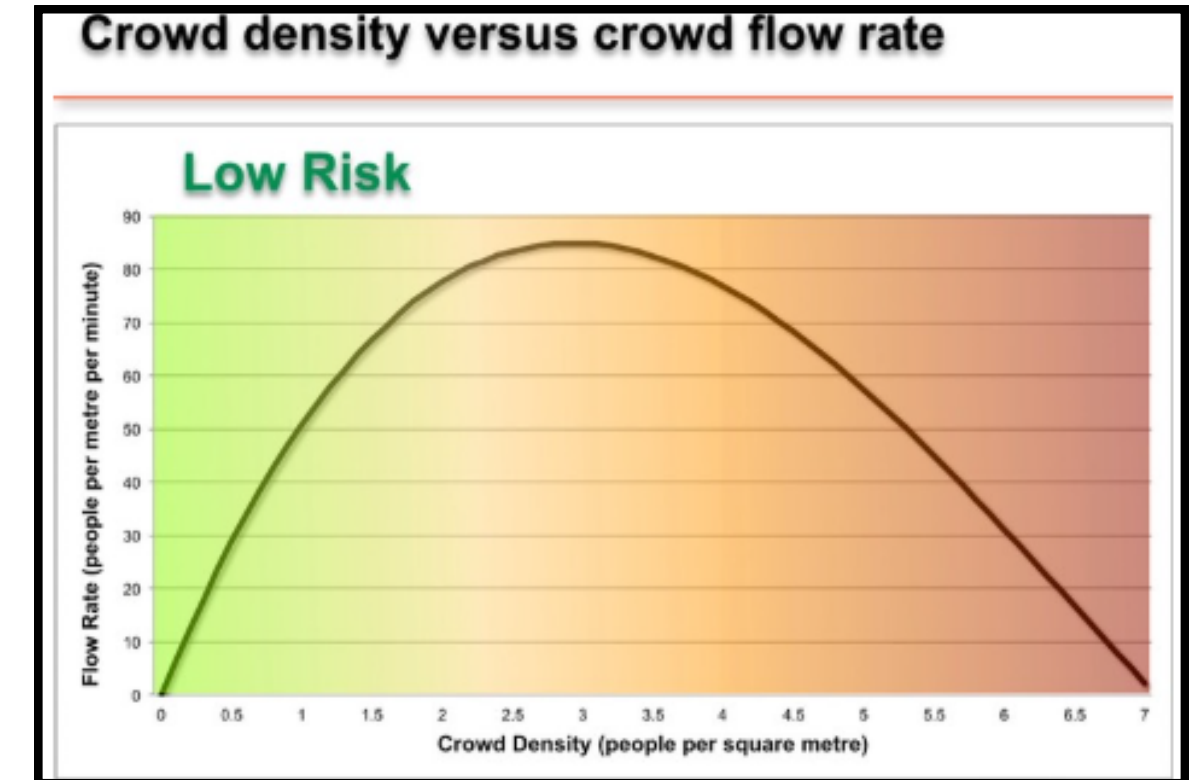
• How will it be able to solve the problem?

1. Detects crowd density per square metre using video feeds
2. Classifies high-risk zones instantly with visual heatmaps
3. Generates real-time AI recommendations for safe crowd movement



• USP of the proposed solution

1. A complete AI-powered solution that detects, predicts, and guides
2. Built for real-world deployment at scale
3. Prevents disasters by enabling timely, data-driven action
4. Designed for large-scale use at events, stations, and public gatherings



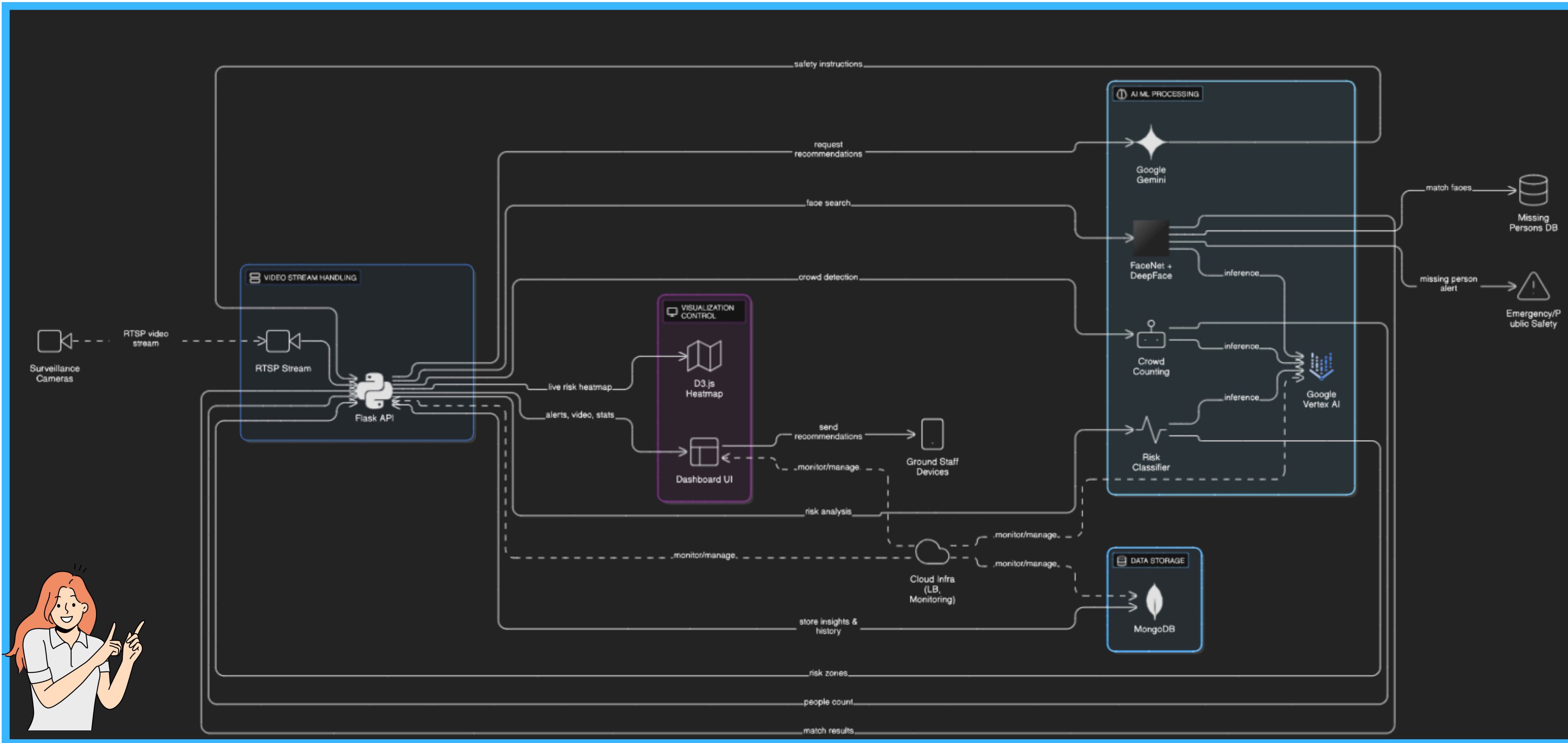
List of features offered by the solution



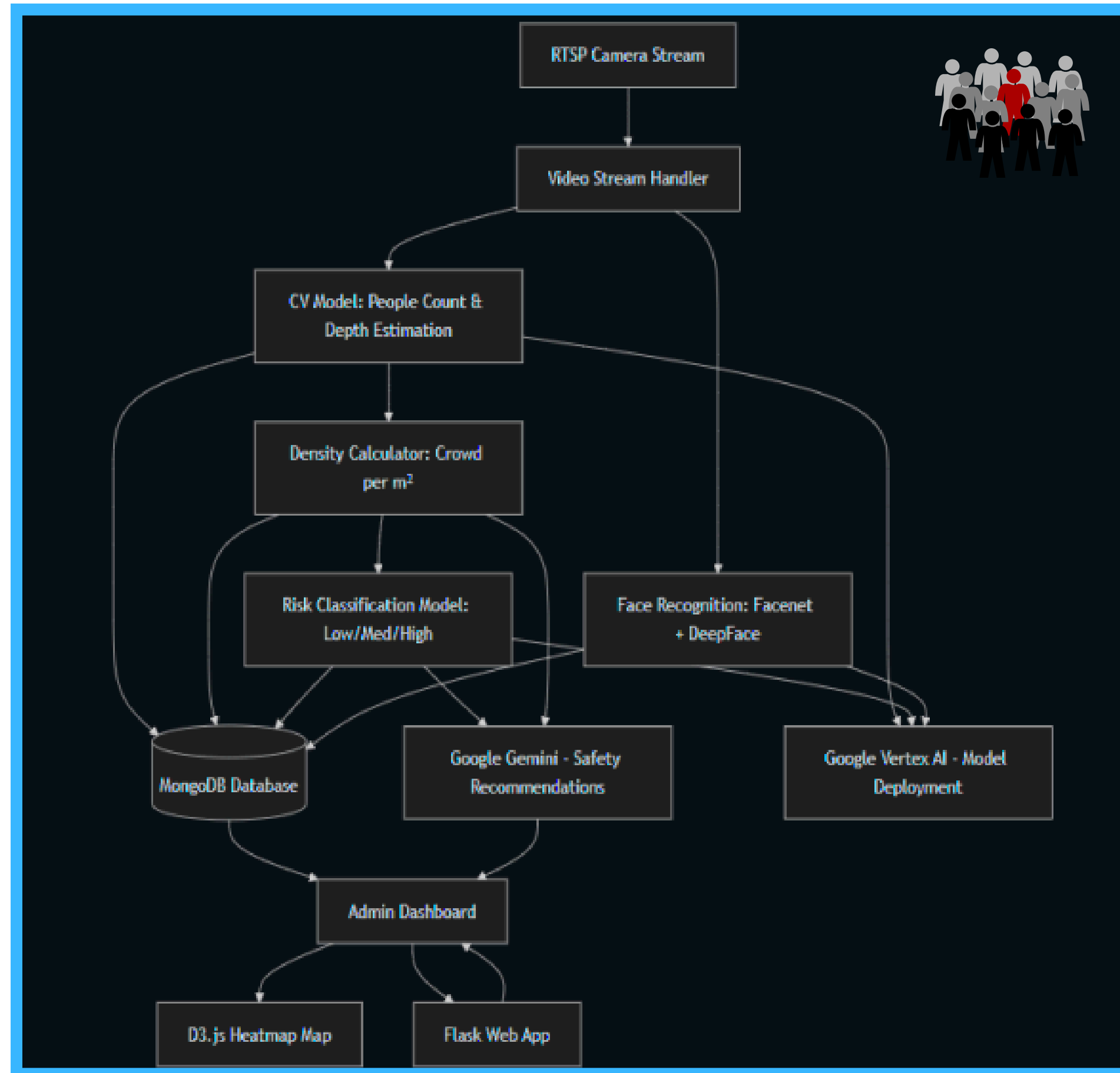
1. **Real-time People Detection** : Helps track crowd size instantly to prevent overload.
2. **Crowd Density Estimation** : Measures how packed an area is to assess safety levels.
3. **Risk Zone (Low/Medium/High)** : Flags areas based on danger levels for quick action.
4. **Live Heatmap Visualization** : Offers a clear visual of crowd pressure for better monitoring.
5. **Facial Recognition for Missing Persons** : Assists in locating individuals quickly in large gatherings.
6. **AI-Generated Crowd Control Instructions** : Guides ground staff with smart, real-time decisions.
7. **Centralized Admin Dashboard** : Enables full system control and alerts from one interface.
8. **RTSP Camera Stream Integration** : Connects easily with existing CCTV systems for live feed.
9. **MongoDB for Data Storage** : Saves crowd insights and history for analysis and reporting.
10. **Scalable with Google Vertex AI** : Supports reliable, cloud-based model deployment at scale.



Process flow diagram or use-case diagram



Process flow diagram or use-case diagram



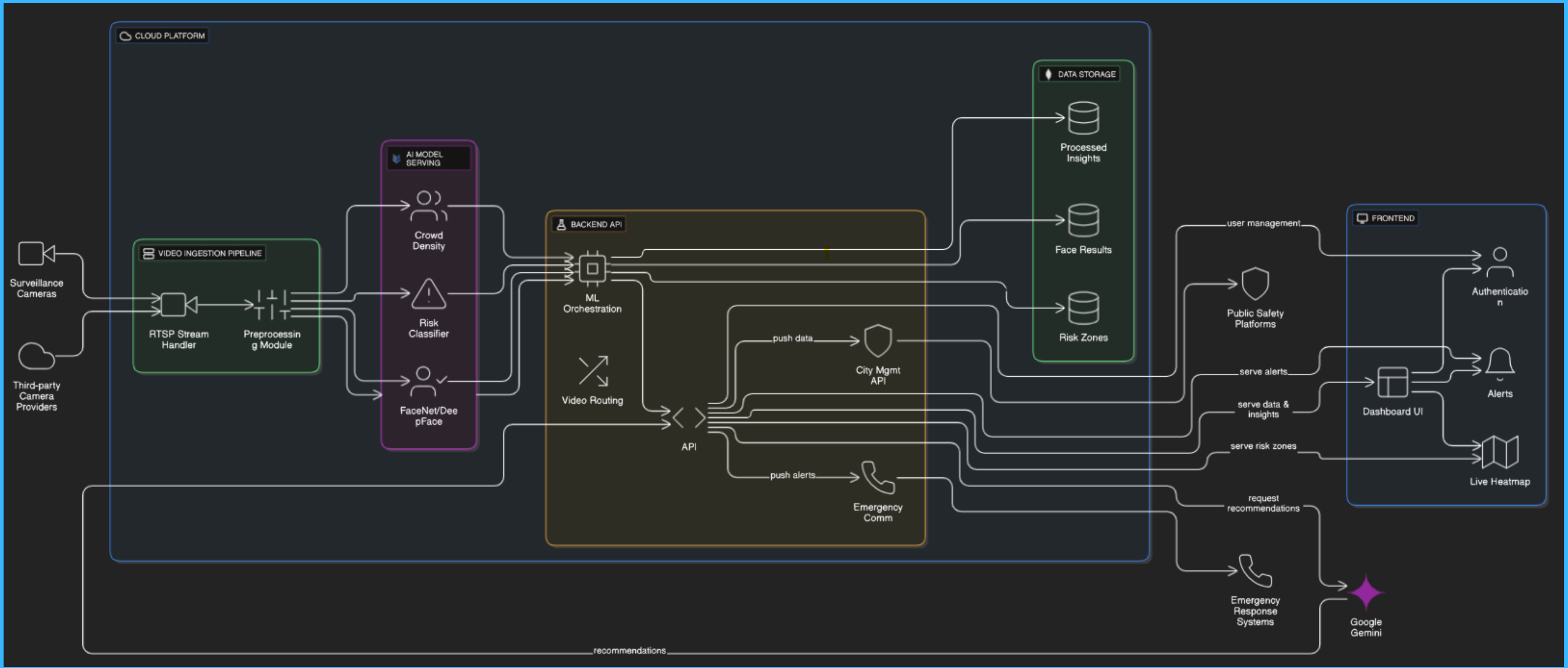
Technologies to be used in the solution :



1. **RTSP Protocol** – For streaming live surveillance camera feeds.
2. **Custom Computer Vision Models** – For real-time people detection and crowd density estimation.
3. **Deep Learning Classifier** – To categorize areas into Low, Medium, and High risk zones.
4. **FaceNet + DeepFace** – For facial recognition and missing person identification.
5. **Google Gemini** – To generate real-time safety instructions for crowd control.
6. **Google Vertex AI** – For scalable model deployment and management in the cloud.
7. **Flask (Python Framework)** – To build backend APIs and serve the dashboard.
8. **MongoDB** – For storing detection results, risk data, and face recognition logs.
9. **D3.js** – For visualizing live heatmaps and crowd density on an interactive map.



Architecture diagram of the proposed solution



Wireframes/Mock diagrams of the proposed solution



Drishti AI

Advanced Crowd Monitoring Command Center

Real-time AI-powered surveillance system for crowd density monitoring, risk zone detection, missing person recognition, and intelligent safety recommendations at large public gatherings.

▶ Launch Command Center

Watch Demo

Powerful Features

Comprehensive AI-powered crowd monitoring capabilities



Real-time Crowd Detection

Advanced AI-powered people counting and density analysis across multiple zones



Missing Person Recognition

Facial recognition technology to identify and locate missing individuals instantly



Risk Zone Monitoring

Intelligent risk assessment with color-coded zones and real-time alerts



AI-Powered Recommendations

Smart crowd control suggestions powered by Google Gemini AI



Multi-Camera Surveillance

Seamless integration with RTSP feeds and multiple camera angles



Predictive Analytics

Historical data analysis and crowd behavior prediction algorithms



Drishti

AI Command Center

11:23:30

 7

Admin User

×

Dashboard

Live Feeds

Risk Analytics

Missing Persons

Safety Recommendations

Logs / History

Command Center Overview

Real-time crowd monitoring and AI-powered analytics

Surveillance Feed – Gate A

Gate A

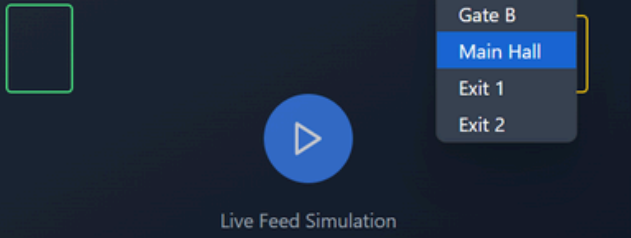
Gate A

Gate B

Main Hall

Exit 1

Exit 2



▶

Live Feed Simulation

People Count: 245 Frame Rate: 33 FPS

Live

Live Crowd Density Heatmap

Main Entrance

Food Court

Exit A

Stage Area

VIP Section



Guard AI

Monitoring

11:12:49 AM

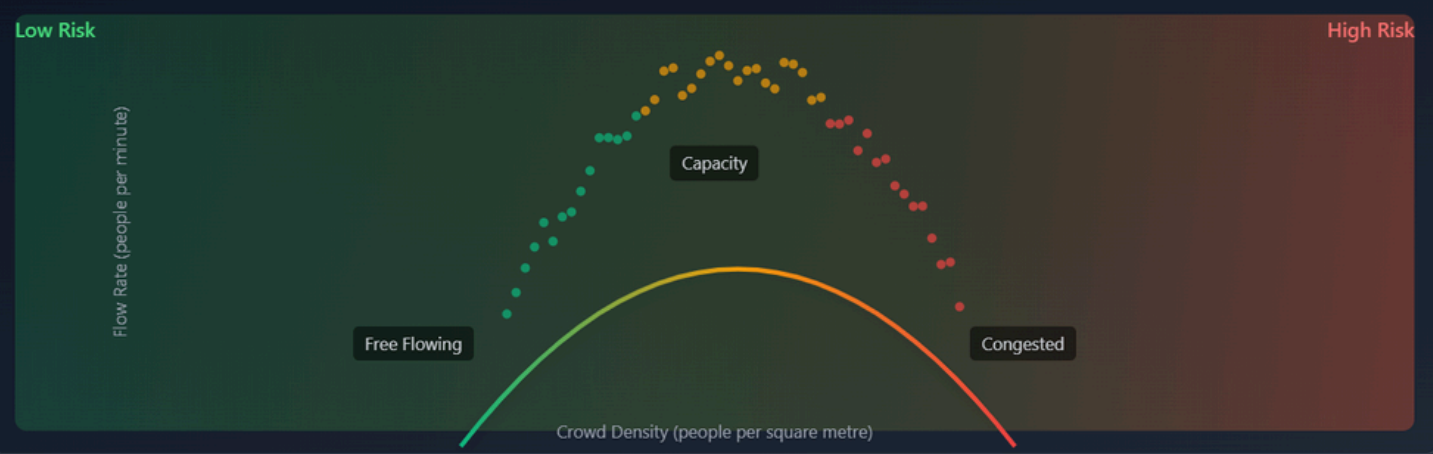
 3

Admin User

Crowd Density vs Flow Rate Analysis

Low Risk

High Risk



Capacity

Free Flowing

Congested

Stable Flow

Density: 0.5-2.0 p/m²

Optimal crowd movement

Unstable Flow

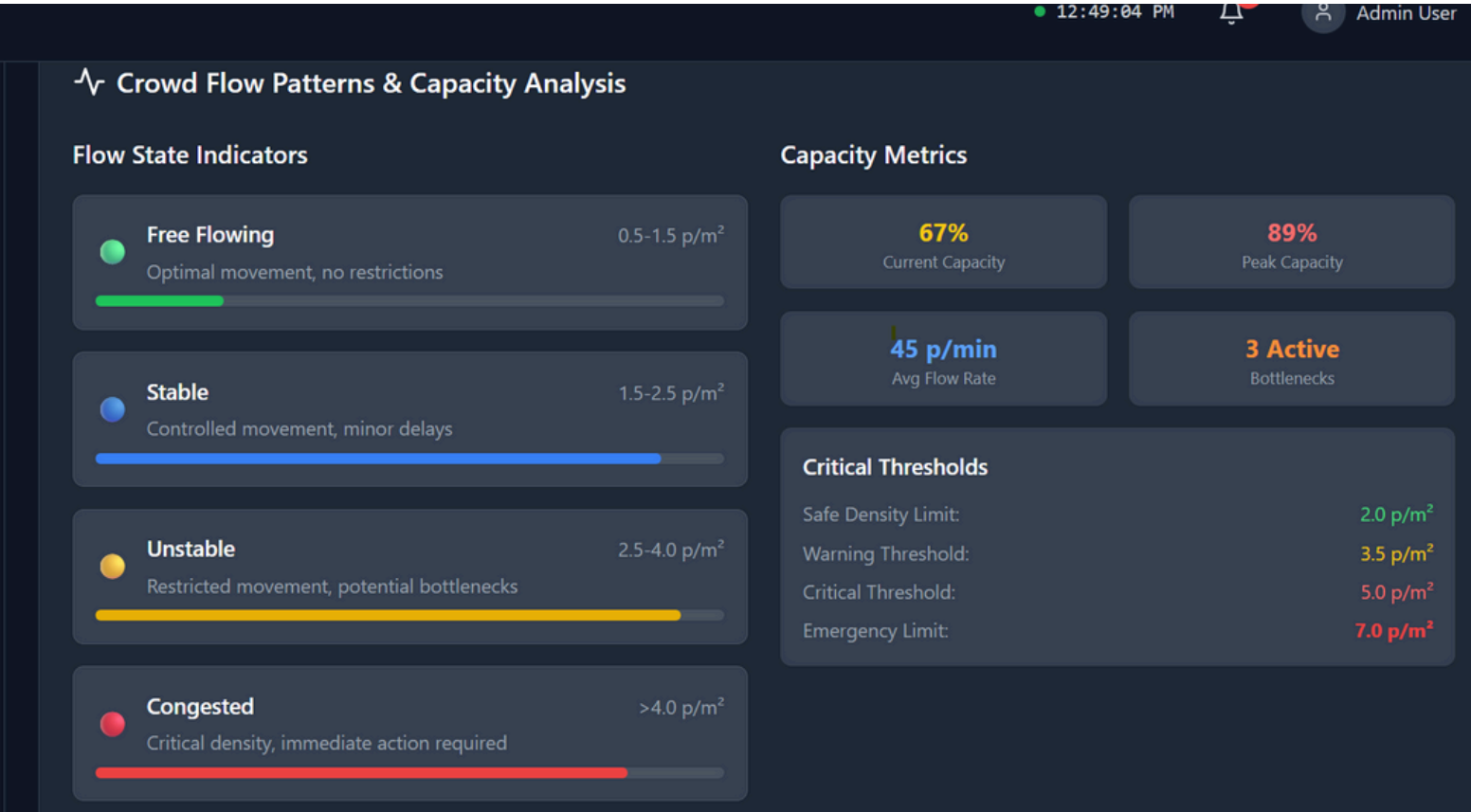
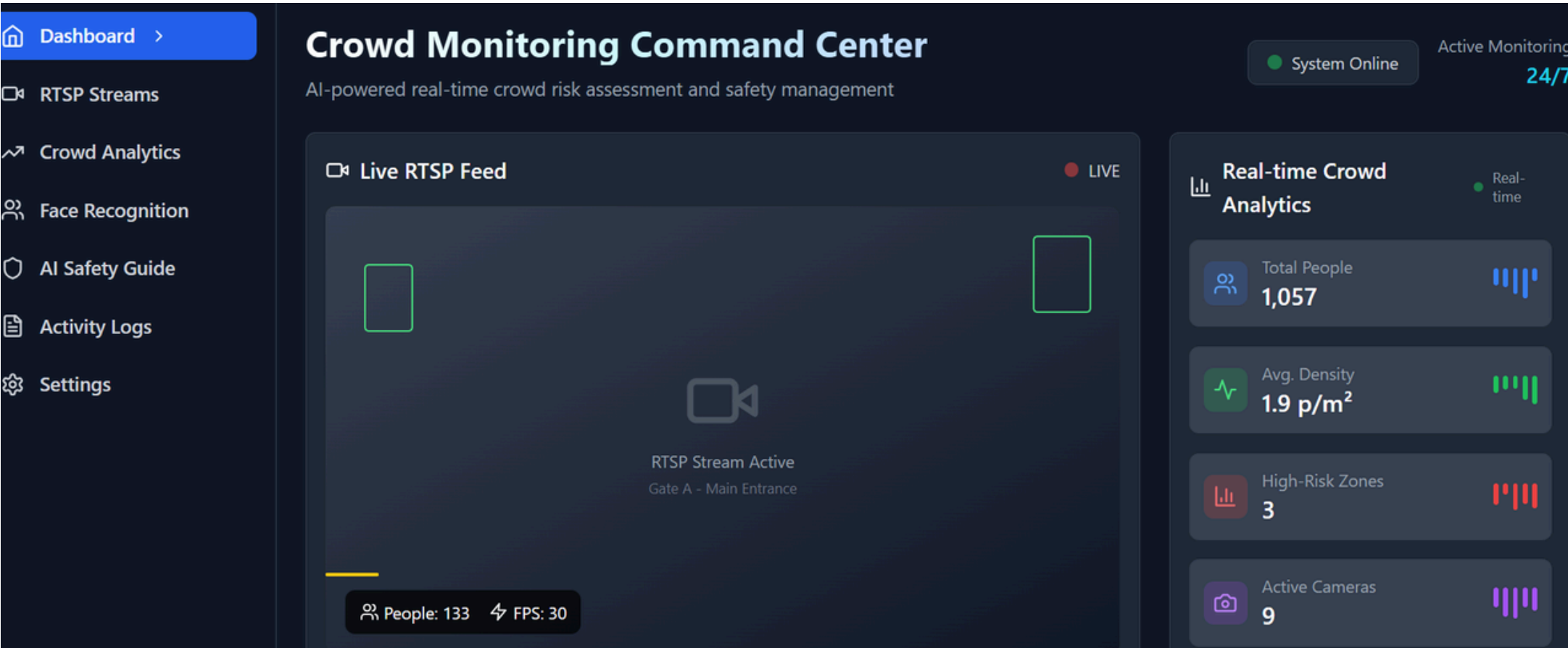
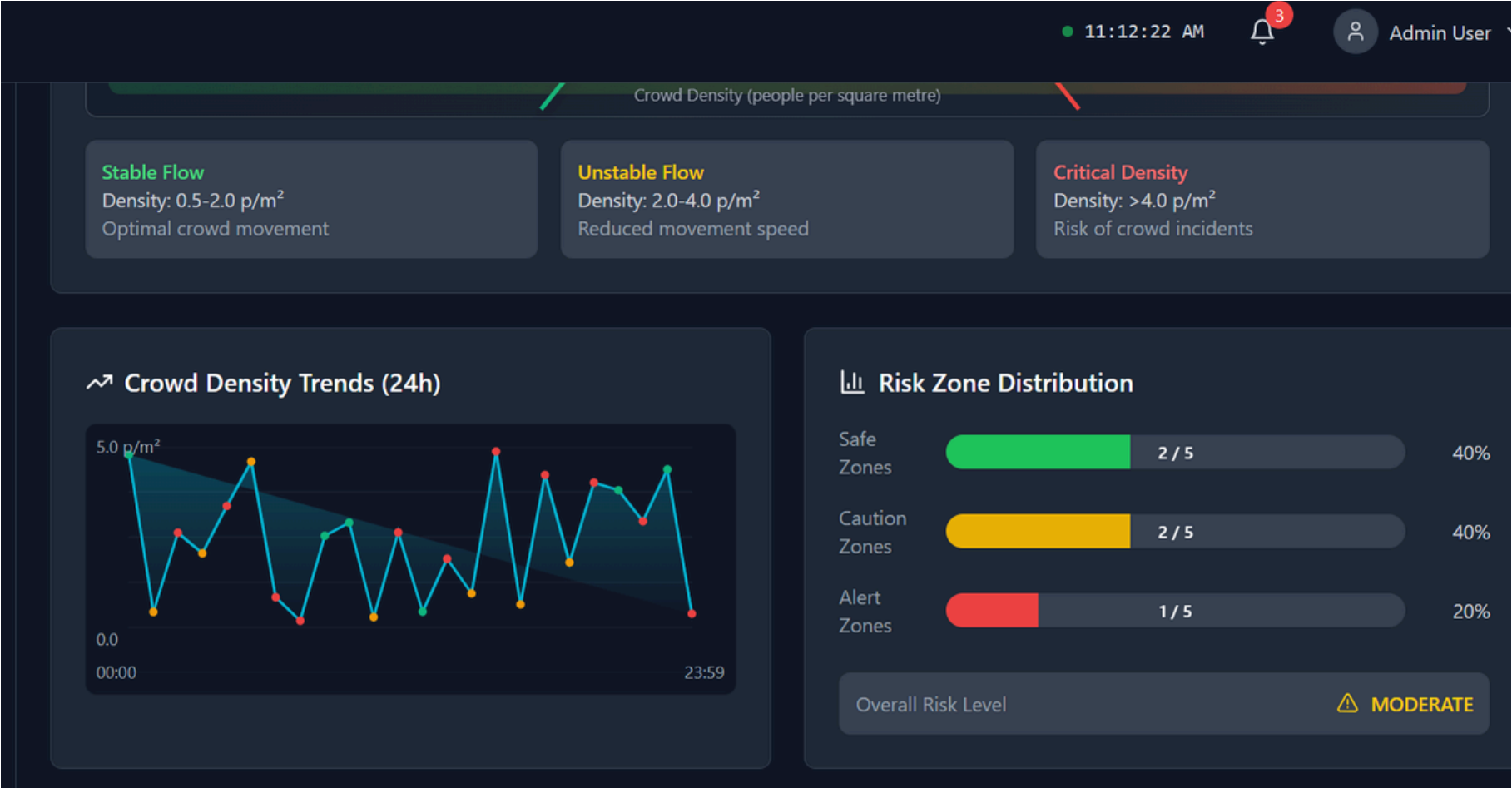
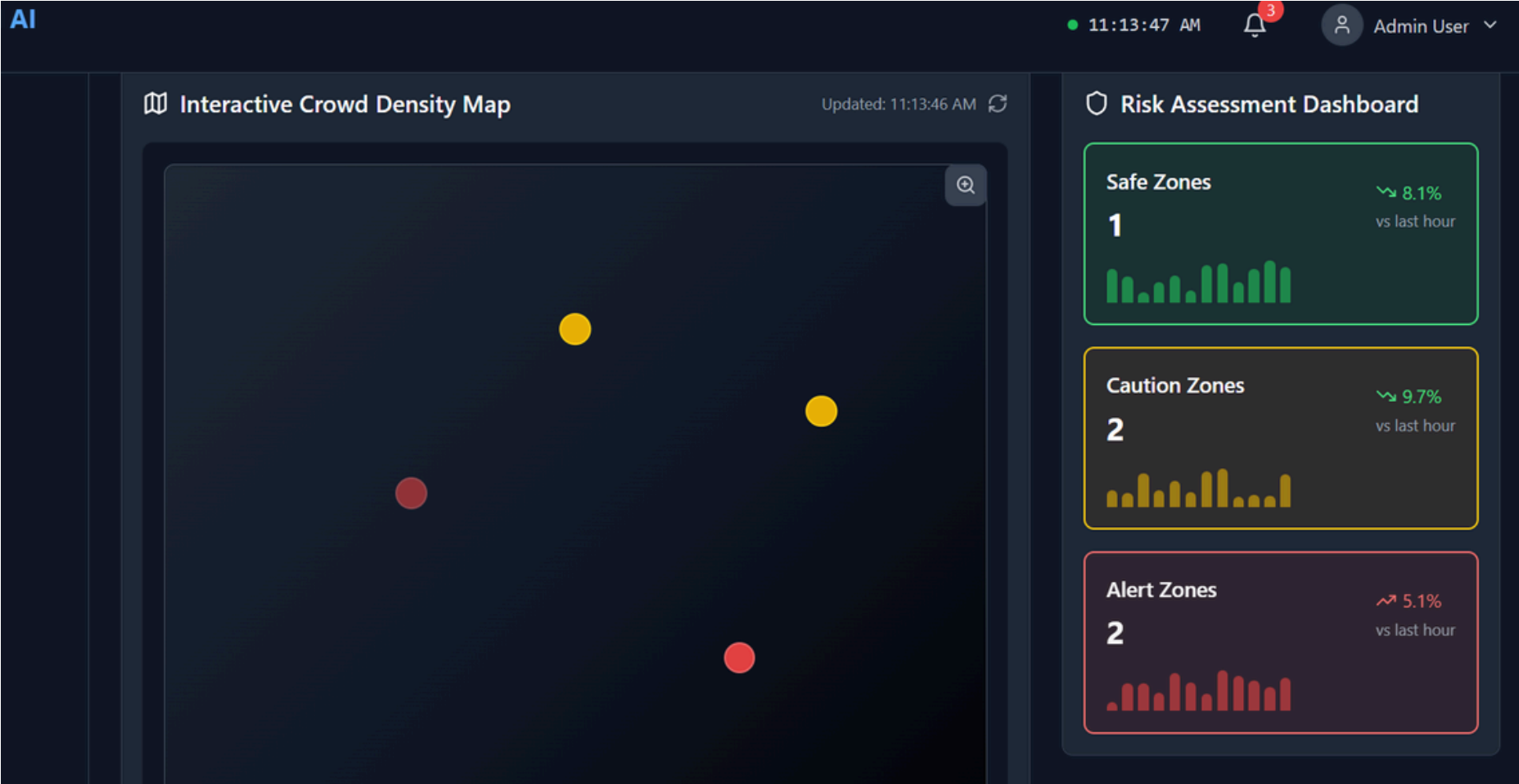
Density: 2.0-4.0 p/m²

Reduced movement speed

Critical Density

Density: >4.0 p/m²

Risk of crowd incidents





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Thank you!

